The smallest affine input in geometric twisting

Human Compression project

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Abstract

Example 5.11 of Piedade–Tranchida, arXiv:2505.24662v1, says that a particular affine quotient at parameter one is not residually connected, and infers that its twist is not regular. The inference fails. We prove from the literal presentations and coset subgroups that the twist is a regular rank-three geometry for every positive parameter, including one. No finite computation is needed for this conclusion.

1 The source construction and its coordinates

Let W have the literal presentation on source page 45:

$$W = \langle a_1, a_2, a_3, a_4 \mid a_i^2 = 1,\ [a_1, a_4] = [a_2, a_3] = 1, (a_1a_2)^3 = (a_1a_3)^3 = (a_2a_4)^3 = (a_3a_4)^3 = 1\rangle.$$

Set $w = a_3a_4a_1a_2$, $N_s = \langle\langle w^{2s}\rangle\rangle$ and $A_s = W/N_s$, for $s \geq 1$. Let p and q permute the four named generators by (12)(34) and (23), respectively. They preserve the defining Coxeter relations. Moreover,

$$p(w) = (a_4a_3)w^{-1}(a_3a_4), \qquad q(w) = (a_3a_2)w(a_2a_3). \tag{1}$$

The first identity is cancellation; the second also uses $a_2a_3 = a_3a_2$. Hence they preserve N_s and induce an action of the abstract diagram group $B = \langle p, q \mid p^2 = q^2 = (pq)^4 = 1\rangle \cong D_8$ on A_s. The inverse permutations ensure equality of the normal closures, rather than just one inclusion. Put $G_s = A_s \rtimes B$. We use left-action coordinates $(a, b)(c, d) = (a\, b(c), bd)$, so $bab^{-1} = b(a)$. For the involutions p, q this agrees with the source's conjugation notation without a choice of inverse action.

In the source take $B_p = \langle q\rangle$, $B_q = \langle p\rangle$, and choose $F_0 = 1$. The orbits of 1 under $1, B_p, B_q, B$ are, respectively, $\{1\}, \{1\}, \{1, 2\}, \{1, 2, 3, 4\}$. Thus its orbit-intersection condition holds. The following are the literal three maximal parabolics of the twisting system, as proved in Lemma 1 rather than inferred from a quotient intersection property:

$$P_L = B, \qquad P_p = \langle a_1\rangle \rtimes \langle q\rangle, \qquad P_q = \langle a_1, a_2\rangle \rtimes \langle p\rangle. \tag{2}$$

In particular these subgroups can be studied even when the input coset system fails residual connectedness. We do not invoke a transfer theorem whose input hypotheses are then absent.

2 Identifying the literal quotient parabolics

We retain the standard faithful affine Coxeter realization of the literal presentation. This external input is needed here to identify the actual quotient intersections in the source construction, as well as for the order calculation and input-failure discussion below. Let $Q = \{v \in \mathbb{Z}^4 : \sum v_i = 0\}$ and let (π, v) act by $z \mapsto \pi z + v$. In $W = Q \rtimes S_4$ take

$$a_1 = ((12), 0),\ a_2 = ((23), 0),\ a_3 = ((14), e_1 - e_4),\ a_4 = ((34), 0).$$

Direct multiplication gives $w^2 = (1, (1, 1, -1, -1))$. Consequently N_s is the lattice sM, where

$$M = 2Q + \mathbb{Z}(1, 1, -1, -1) = \{v \in Q : v_i \equiv v_j \pmod 2 \text{ for every } i, j\}.$$

Indeed conjugation permutes the coordinates of the translation; differences of orbit vectors generate $2Q$, and every orbit vector is congruent to $(1, 1, -1, -1)$ modulo $2Q$. Thus $[Q : M] = 4$, $|A_s| = 96s^3$ and $|G_s| = 768s^3$. The affine model is an explicit retained premise of the literal-source identification. The subsequent abstract marking argument uses only the source presentation and an elementary S_4 quotient.

Lemma 1. *For every $s \geq 1$, the actual quotient maximal parabolics satisfy*

$$(A_s)_3 \cap (A_s)_4 = \langle a_1, a_2 \rangle, \quad (A_s)_2 \cap (A_s)_3 \cap (A_s)_4 = \langle a_1 \rangle, \quad \bigcap_{i=1}^{4} (A_s)_i = 1.$$

Consequently the literal twisting subgroups are exactly (2).

Proof. The four maximal parabolics of W fix the points

$$c_1 = (3, -1, -1, -1)/4, \quad c_2 = (1, 1, -1, -1)/2, \quad c_3 = 0, \quad c_4 = (1, 1, 1, -3)/4.$$

Each has linear part S_4: its three generating transpositions form an edge tree on the four coordinates. An element fixing c_i has translation $c_i - \pi c_i$. The quotient embeds each such parabolic, since a pure translation fixing a point is the identity.

The third parabolic is the pure linear S_4. Its element π is in the image of the fourth precisely when $c_4 - \pi c_4 \in sM$. A nonzero such difference is a vector with exactly two nonzero coordinates, one 1 and one -1. It cannot belong to sM: for $s \geq 2$ divisibility fails, and for $s = 1$ its four coordinate parities are not equal. Thus π must fix coordinate 4. These elements form exactly $\langle (12), (23) \rangle = \langle a_1, a_2 \rangle$.

Within this subgroup, membership in the second parabolic means $c_2 - \pi c_2 \in sM$. Such a difference has all coordinates in $\{-1, 0, 1\}$. It is zero exactly when π preserves both blocks $\{1, 2\}, \{3, 4\}$. The only other way to have all equal coordinate parities is to interchange the blocks, which is impossible when coordinate 4 is fixed. Thus π preserves the blocks and fixes both 3 and 4, leaving only $\langle (12) \rangle$. Finally membership in the first parabolic forces coordinate 1 to be fixed, by the same two-odd-coordinates argument for $c_1 - \pi c_1$. This leaves the identity, proving all three formulas. □

3 A uniform direct proof of regularity

Theorem 2. *For every $s \geq 1$, the literal coset system $(G_s, (P_L, P_p, P_q))$ is a thin, residually connected, flag-transitive rank-three geometry. Its distinguished marking is (x, p, q) with $x = a_1$, and its adjacent product orders are six and four. In particular it is a regular hypertope at $s = 1$.*

Proof. The action of B on the four names is transitive. Therefore x, p, q generate G_s. The pair a_1, a_2 generates an exact S_3 in every A_s. Its order is at most six by its two involutions and braid of length three. For the lower bound use the quotient map

$$a_1 \mapsto (12), \quad a_2 \mapsto (23), \quad a_3 \mapsto (14), \quad a_4 \mapsto (34) \quad \text{in } S_4.$$

All defining relations hold and the image of w is $(12)(34)$, so the map kills w^{2s} for every s. The first two images generate S_3, proving the lower bound and the claimed injection. The same S_4 image distinguishes all four named generators in A_s. The diagram group D_8 permutes these names faithfully, so its action on A_s is faithful as well. Thus our abstract semidirect factor is exactly the source's actual group of diagram automorphisms.

Conjugation by p interchanges a_1, a_2, whence

$$H_2 = \langle x, p\rangle = \langle a_1, a_2\rangle \rtimes \langle p\rangle \quad \text{has order 12.}$$

Its involutory generators have product of order six: $(xp)^2 = a_1a_2$ has order three and the B projection of xp has order two. Also $H_0 = \langle p, q\rangle = B$ has order eight, and $H_1 = \langle x, q\rangle \cong C_2^2$, since q fixes a_1 and x is nontrivial in the base. Semidirect coordinates give

$$H_0 \cap H_2 = \langle p\rangle, \qquad H_0 \cap H_1 = \langle q\rangle, \qquad H_1 \cap H_2 = \langle x\rangle. \tag{3}$$

For the last equality the projections lie in $\langle q\rangle \cap \langle p\rangle = 1$; the base part of H_1 is exactly $\langle x\rangle$. The other two equalities are immediate from the two-factor coordinates. All named generators are distinct, so these identities include the complete intersection property for the three-generator marking, with Borel subgroup one. By (2) these are precisely the source's maximal parabolics, reordered as (H_0, H_1, H_2).

We verify flag transitivity directly. Take three pairwise incident cosets, one of each type. Translate a common representative of the H_0 and H_2 cosets to the identity. The remaining coset gH_1 meets H_0 at some u and H_2 at some v. Then $v^{-1}u \in H_1$, so $v^{-1}u = x^{\epsilon}q^{\delta}$. The element $vx^{\epsilon} = uq^{-\delta}$ lies in all three cosets. Thus every flag has a common representative, is a translate of a standard flag, and extends to a chamber. Each rank-two residue has acting group one of H_0, H_1, H_2, and its two child groups in (3) generate it. Its incidence graph is connected: a product of parabolic elements gives a path from the standard flag. The whole incidence graph is connected because the maximal parabolics generate G_s. This proves residual connectedness. Finally every minimal parabolic has order two and the Borel is one, so all panels have size two. The regular chamber action is faithful; an automorphism fixing a chamber fixes each adjacent chamber in a thin geometry and then all chambers by connectedness. Thus it is also the full type-preserving automorphism group. □

4 The input failure and the finite model

The four maximal proper parabolics of W fix, respectively,

$$c_1 = (3, -1, -1, -1)/4, \quad c_2 = (1, 1, -1, -1)/2, \quad c_3 = 0, \quad c_4 = (1, 1, 1, -3)/4.$$

Their linear parts are S_4, and an element fixing c_i has translation $c_i - \pi c_i$. At $s = 1$, the intersection of the c_2 and 0 stabilizers modulo M is the order-eight group of linear permutations preserving or interchanging the blocks $\{1, 2\}, \{3, 4\}$. This follows because $c_2 - \pi c_2 \in M$ exactly in those two cases. Intersecting additionally with the c_1 stabilizer forces coordinate 1 to be fixed, leaving $\langle(34)\rangle$; intersecting with the c_4 stabilizer forces coordinate 4 to be fixed, leaving $\langle(12)\rangle$. Nonzero differences $c_1 - \pi c_1$ and $c_4 - \pi c_4$ have exactly two odd coordinates and hence cannot lie in M. These two child groups generate only a group of order four, so the rank-two residue of the flag of types 2 and 3 is disconnected. The direct proof above shows that this input defect disappears in the particular twisting construction.

The supplementary integer program constructs all four quotient maximal parabolics and their actual intersections, and reconstructs the order-96 quotient, the eight literal diagram automorphisms and their composition, and the order-768 semidirect product. It compares all three literal parabolics, all 64 generated-subgroup intersections and 343 proper product identities. The full helper is retained alongside it; none of its other finite parameter checks is a premise of the theorem. There is no Lean claim. The exact source assertion being corrected is the page-46 conclusion that the $s = 1$ twist is not a regular hypertope. This is a version-bound theorem/example correction, not a numbered-conjecture solution, a full-source admission or a claim of historical priority.